

NIC CPD Micro-Credential: Phlebotomy & Specimen Collection for Professional Caregivers

Issuing organisation: National Institute of Caregivers Nigeria (NIC)

Credential type: Stand-alone Continuing Professional Development (CPD) Micro-Credential

Online duration: 5–6 CPD learning hours

Practical component: Separate supervised competency verification where the learner's role, training, employer and applicable requirements permit venepuncture or capillary sampling

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Essential boundary: Completion of this online CPD does **not** establish practical competence to perform venepuncture, capillary sampling or any other invasive procedure. It is not a licence to practise phlebotomy. A practical endorsement may be issued only after approved hands-on education and supervised assessment by an appropriately qualified and authorised assessor.

1. Course title

NIC CPD Micro-Credential: Phlebotomy & Specimen Collection for Professional Caregivers

2. Course overview

This evidence-informed CPD introduces professional caregivers, healthcare assistants and related support personnel to the purpose, preparation, safety, collection, handling, labelling, transport and documentation of blood specimens.

The course is designed around a simple quality principle:

The right patient + right specimen + right container + right label + right handling = reliable laboratory result.

Learners study phlebotomy as a coordinated patient-safety and laboratory-quality process. They learn what the procedure involves, why each stage matters, how to recognise risks and complications, and when to stop, seek supervision or refer to an authorised professional.

The course follows the WHO guidance on drawing blood and safe phlebotomy ¹, current laboratory-quality principles, CDC blood-culture and exposure resources ² ³, and current CLSI order-of-draw resources ⁴. Tube colours, order of draw, storage, transport and device instructions vary by laboratory and manufacturer; NIC must verify the local laboratory's current SOP before publication.

3. Target audience

The target audience includes professional caregivers, healthcare assistants, nursing assistants, home-care workers, care-facility support staff, hospital support personnel, community-care workers and existing care professionals seeking CPD development.

A learner's actual authority to collect specimens depends on role, training, employer and facility policy, supervision, applicable Nigerian requirements and professional standards. The course must not imply that employability interest alone authorises invasive practice.

4. Entry expectations

Learners should understand basic caregiving, communication, infection prevention, confidentiality and escalation. No clinical qualification is required for the online knowledge pathway. Practical verification requires suitable equipment, a safe training environment, an authorised assessor and a written scope of practice.

5. CPD learning hours

The online component is designed for **5–6 CPD learning hours**, including lessons, diagrams, knowledge checks, scenarios, module assessments, final assessment and downloadable job aids. Practical competency verification is separate.

6. Central competency statement

Explain and support safe, accurate and professional blood specimen collection and handling within the learner's authorised scope of practice, while protecting patient safety, specimen integrity, infection prevention, dignity and confidentiality.

7. Learning outcomes

By completion, learners should be able to:

1. Explain the purpose and importance of phlebotomy.
2. Describe common reasons blood specimens are collected.

3. Explain caregiver role, accountability, delegation and scope.
4. Identify common blood specimen types and collection requirements.
5. Explain basic anatomy relevant to venepuncture.
6. Identify equipment and preparation requirements.
7. Apply patient identification, consent and communication principles.
8. Explain infection prevention, PPE and sharps safety.
9. Describe the general venepuncture sequence without unsupervised practice.
10. Explain common specimen tubes, additives and uses.
11. Explain the principle of order of draw and additive carryover.
12. Describe capillary blood-sampling principles.
13. Explain bedside labelling and documentation.
14. Explain packaging, storage and transport requirements.
15. Identify specimen rejection causes.
16. Recognise common complications and adverse reactions.
17. Explain immediate caregiver responses to complications.
18. Protect dignity, privacy, communication and confidentiality.
19. Identify when to stop, escalate or refer.
20. Explain when supervised practical education is required.

8. Course structure

Module	Title	Lessons	Assessment
1	Foundations, Purpose and Scope	1. What Phlebotomy Is; 2. Why Specimens Matter; 3. Role, Delegation and Accountability	6 questions
2	Patient Safety, Identification, Consent and Communication	4. Identification; 5. Consent and Dignity; 6. Communication and Vulnerability	6 questions
3	Anatomy, Equipment, Preparation and Infection Prevention	7. Basic Anatomy; 8. Equipment and Preparation; 9.	6 questions

		Infection Prevention and Sharps	
4	Venepuncture Principles and Practical Boundaries	10. General Sequence; 11. Collection Systems and Site Awareness; 12. Complications and Stop Rules	6 questions
5	Tubes, Additives, Order of Draw and Capillary Sampling	13. Tubes and Additives; 14. Order of Draw; 15. Capillary Sampling and Special Populations	6 questions
6	Specimen Integrity, Labelling, Handling and Transport	16. Labelling and Documentation; 17. Rejection and Quality; 18. Packaging, Storage and Transport	6 questions
7	Integrated Practice, Reporting and Competency Pathways	19. Special Clinical Considerations; 20. Adverse Events and Exposure; 21. Integrated Judgement and Verification	6 questions

9. Complete Module 1 — Foundations, Purpose and Scope

Lesson 1. What Phlebotomy Is

Phlebotomy is the collection of blood for laboratory testing. It may involve venous blood, capillary blood or, in specialist contexts, other authorised methods. This CPD is not a licence to perform any of them.

The caregiver must know the difference between **learning about phlebotomy** and **being authorised and competent to perform it**. A learner may support preparation, identification, reassurance, observation and documentation without being authorised to insert a needle.

Scenario: A caregiver completes this course and is asked to draw blood at home. **Best response:** Check written scope, training, employer policy and supervision. If not

authorised, do not perform the procedure; refer to an authorised professional.

Knowledge check: What does phlebotomy involve? **Answer:** Blood collection for laboratory testing under appropriate authority.

Lesson 2. Why Specimens Matter

Blood tests support diagnosis, monitoring, treatment decisions and screening. A wrong patient, wrong tube, haemolysed sample, insufficient volume or delayed specimen can produce unreliable or rejected results.

Laboratory quality begins before the specimen reaches the laboratory. Patient identification, preparation, correct container, correct handling and timely transport are all patient-safety activities.

Scenario: A specimen is collected correctly but labelled with another person's details.

Best response: Treat as a serious identification error, stop the process and follow the facility's incident and laboratory policy.

Knowledge check: Why does specimen integrity matter? **Answer:** The result may be wrong, delayed or unsafe to use.

Lesson 3. Role, Delegation and Accountability

Authority is not created by confidence or a family request. It depends on role, training, employer/facility policy, supervision and applicable requirements. A caregiver must know what they may do, what they may assist with and what requires referral.

Delegation does not remove accountability. The person delegating must ensure competence and supervision; the learner must refuse a task outside training or authorisation and report the boundary concern.

Scenario: A senior worker says, "Everyone does this here." **Best response:** Request documented authorisation and appropriate supervision; do not proceed on informal instruction.

Knowledge check: Can online completion authorise venepuncture? **Answer:** No.

10. Complete Module 2 — Patient Safety, Identification, Consent and Communication

Lesson 4. Patient Identification

Use the approved identifiers and match the patient to the request/order and specimen labels. Do not rely solely on room number, bed number, appearance or recognition. If identification does not match, stop and resolve the discrepancy through the authorised process.

Identification should occur before collection and again at labelling where the policy requires bedside identification. Never borrow a label, pre-label an unidentified tube or guess details.

Scenario: The request says “Mr Ade,” but the wristband and verbal identifiers do not match. **Best response:** Stop and escalate the mismatch.

Lesson 5. Consent and Dignity

Explain what will happen in language the person can understand, answer questions within competence, seek appropriate consent and respect refusal. A caregiver must not use threat, shame or deception. Concerns about capacity or communication should be escalated through the authorised pathway.

Maintain privacy, appropriate draping, cultural sensitivity and gender considerations where relevant. Do not promise a result or suggest that collection guarantees a diagnosis.

Scenario: A patient refuses after explanation. **Best response:** Respect refusal, inform the authorised professional and document according to policy.

Lesson 6. Communication and Vulnerability

Introduce yourself, confirm preferred communication, explain each stage and reassure without false promises. Children, older adults, people with disabilities, anxious patients and people with limited language access may need extra time, an interpreter or a support person according to policy.

Do not speak over the patient or assume a carer can consent for an adult without proper authority. Protect confidentiality in homes, wards and shared spaces.

Scenario: A relative answers every question for a competent adult. **Best response:** Address the patient respectfully and use the authorised communication/consent process.

11. Complete Module 3 — Anatomy, Equipment, Preparation and Infection Prevention

Lesson 7. Basic Anatomy

Veins return blood to the heart; arteries carry blood away; capillaries are very small vessels. Common venepuncture awareness includes the median cubital, cephalic and basilic veins. Site selection is a trained clinical decision. Some sites require additional caution, and areas with injury, infection, swelling, fistula, IV access or other risk may need avoidance under policy.

This lesson is conceptual. It must not encourage a learner to search for a vein or insert a needle unsupervised.

Scenario: A learner wants to use an area near an IV line because it is visible. **Best response:** Stop and ask an authorised professional to select an appropriate site.

Lesson 8. Equipment and Preparation

Common supplies may include gloves, tourniquet, approved skin preparation, needles, holders, syringes where applicable, collection tubes, gauze, dressing, sharps container, labels, request forms, specimen bags and transport containers. Exact equipment varies by facility and laboratory.

Preparation includes hand hygiene, PPE, checking packaging and expiry, equipment integrity, lighting, patient positioning, sharps disposal readiness, instructions and local SOP. The sharps container should be available before a needle is used.

Scenario: The only available needle packet is damaged. **Best response:** Do not use it; obtain intact approved equipment.

Lesson 9. Infection Prevention and Sharps Safety

Apply standard precautions to blood and body fluids. Perform hand hygiene, use appropriate gloves/PPE, use single-use equipment, prepare the skin according to current policy, prevent cross-contamination and dispose of sharps immediately in an approved sharps container.

Never recap a used needle unless a specific approved safety device/procedure requires it and the relevant policy explicitly permits it. Use safety-engineered devices where available. Needlestick injuries and blood exposure require immediate first aid, urgent reporting and occupational-health/emergency follow-up according to policy.

Scenario: A colleague asks you to recap a needle to carry it to the bin. **Best response:** Do not recap; use the sharps container at the point of use and report unsafe practice.

12. Complete Module 4 — Venepuncture Principles and Practical Boundaries

Lesson 10. The General Venepuncture Sequence

The conceptual sequence is:

Prepare → Identify → Explain → Consent → Position → Hand Hygiene/PPE → Select Site → Apply Tourniquet → Prepare Site → Collect Specimen → Release Tourniquet → Remove Needle → Apply Pressure → Dispose of Sharps → Label → Document → Transport.

Each stage protects the patient, collector and specimen. The sequence is a learning map, not permission for unsupervised practice. Needle insertion, tube filling, pressure and device operation must be taught in person where authorised.

Scenario: A learner wants to practise on a family member using a home kit. **Best response:** Do not practise on a real person outside approved training.

Lesson 11. Collection Systems and Site Awareness

Collection systems may include evacuated tubes, holders, needles, syringes where applicable and capillary devices. Follow the approved system and manufacturer instructions. Do not mix systems or improvise components.

Site selection, tourniquet use, skin preparation, tube filling, mixing, release, removal and pressure are technical skills. The online learner should understand their purpose and risks but must not treat diagrams as a substitute for supervised practice.

Lesson 12. Complications and Stop Rules

Possible complications include pain, bruising/haematoma, bleeding, dizziness, fainting, vasovagal reaction, nerve-related symptoms, accidental arterial puncture awareness, infection risk, needlestick injury, persistent bleeding and patient distress.

Use the response sequence:

Stop → Protect the patient → Apply appropriate immediate measures within training → Escalate → Document.

Stop and escalate for severe pain, electric-shock sensation, sudden swelling, bright pulsatile bleeding, persistent bleeding, fainting, altered responsiveness, patient refusal, equipment failure, wrong identification, contamination or any uncertainty.

13. Complete Module 5 — Tubes, Additives, Order of Draw and Capillary Sampling

Lesson 13. Tubes and Additives

Tube colour is not universal. The laboratory's current tube guide and manufacturer instructions take priority. Common categories include:

Category	Common additive/purpose	Typical awareness use
Blood culture bottles/tubes	Culture media	Microbiology culture, with strict collection requirements
Citrate/light blue	Sodium citrate	Coagulation testing; correct blood-to-additive ratio matters
Serum/red or gold	No anticoagulant, clot activator and/or gel	Serum chemistry/serology, according to laboratory guide
Heparin/green	Heparin	Plasma chemistry or specialised testing, according to guide
EDTA/lavender or pink	EDTA	Haematology or blood bank testing, according to guide
Fluoride/oxalate/grey	Glycolysis inhibitor and anticoagulant	Selected glucose/lactate testing, according to guide

Filling, mixing, clotting time, centrifugation and storage depend on the test and tube. Incorrect tube or additive carryover can invalidate results.

Lesson 14. Order of Draw

Order of draw reduces additive carryover from one tube into another. A commonly used sequence is blood culture, citrate, serum, heparin, EDTA and fluoride/oxalate, but the exact sequence must be verified against the current laboratory SOP, CLSI guidance and manufacturer instructions before publication ⁴.

Do not rely on colour memory. Different systems, special tubes and local procedures may change the required sequence. If the wrong tube is used, do not silently transfer blood; follow laboratory policy and seek guidance.

Lesson 15. Capillary Sampling and Special Populations

Capillary sampling uses a small skin puncture for selected tests. Site selection, skin preparation, approved device, first-drop policy, avoiding excessive squeezing, safe disposal, labelling and documentation must follow current training and device instructions.

Older adults may have fragile skin, fragile veins, bruising, dehydration or communication difficulties. Children require age-appropriate communication, appropriate equipment, consent/guardian procedures and safeguarding. People with disabilities may need communication and positioning adaptation. Difficult access should not become uncontrolled repeated attempts; follow the attempt limit and escalation policy.

14. Complete Module 6 — Specimen Integrity, Labelling, Handling and Transport

Lesson 16. Labelling and Documentation

Label according to policy, preferably at the patient's side where bedside identification is required. Include approved identifiers, date/time, specimen type, collector identification and other required details. Match the label to the patient and request before leaving.

Document collection, unsuccessful attempts, refusal, complications, equipment or identification problems, actions, escalation, collector identity and transport. Use objective language and protect confidentiality.

Scenario: A collector plans to label several tubes at the desk. **Best response:** Follow bedside labelling policy; do not label away from the patient when the policy requires bedside verification.

Lesson 17. Rejection and Quality

Laboratories may reject specimens because of wrong patient, mislabelling, wrong tube, insufficient volume, clotting, haemolysis, contamination, leakage, delay, incorrect storage or unsuitable condition. A rejected specimen can delay care and cause repeat collection.

If an error is discovered, do not conceal or correct the label informally. Quarantine or manage the specimen according to policy, inform the authorised professional/laboratory and complete incident reporting where required.

Lesson 18. Packaging, Storage and Transport

The specimen journey is:

Collection → Identification → Labelling → Documentation → Packaging → Storage → Transport → Laboratory Reception.

Use the approved leak-resistant specimen bag, request-form compartment, transport container and temperature/time requirements. Do not place forms in a contaminated compartment. Follow the laboratory's test-specific stability and transport instructions. Delayed or inappropriate transport may invalidate the result.

15. Complete Module 7 — Integrated Practice, Reporting and Competency Pathways

Lesson 19. Special Clinical Considerations

Additional caution or professional direction may be required for anticoagulant therapy, bleeding disorders, mastectomy or lymph-node procedures, IV lines, fistulas, infection or injury at the site, burns, oedema, severe dehydration, altered consciousness or difficult access.

The caregiver should not independently decide that a risk is acceptable. Stop, explain, protect dignity and refer to an authorised professional.

Lesson 20. Adverse Events and Exposure

For pain, bruising, bleeding, dizziness, fainting, nerve symptoms, suspected arterial puncture, infection concern or persistent bleeding: stop, protect, apply trained immediate measures, call the appropriate professional and document.

For needlestick or blood exposure: perform immediate first aid according to policy, report urgently, seek occupational-health/emergency assessment and preserve relevant information. Do not delay reporting while completing routine paperwork.

Lesson 21. Integrated Judgement and Verification

The integrated process is: **right person, right request, right consent, right equipment, right specimen, right container, right label, right handling, right transport and right handover.**

The online pathway proves knowledge. The practical pathway requires supervised demonstration of the authorised skills, including identification, consent, equipment, infection prevention, technique, complication response, labelling, documentation and transport. No learner should practise on a real person outside approved training.

16. Visual learning plan

Visual	Purpose
Specimen-quality pathway	Show patient-to-laboratory journey
Scope-of-practice decision tree	Show online learning versus authorised practice
Patient-identification card	Show identifiers and mismatch stop rule
Equipment layout	Show preparation without encouraging unsupervised use
Basic vessel diagram	Explain veins, arteries and capillaries
Conceptual venepuncture sequence	Show workflow, labelled “supervised practical skill”
Tube/additive table	Support recognition, with local-SOP warning
Order-of-draw diagram	Show principle and local verification warning
Specimen-rejection matrix	Connect error to patient impact
Complication stop card	Stop → Protect → Immediate measure → Escalate → Document
Specimen transport diagram	Show packaging, storage and handover

Visuals should feature diverse Nigerian adults, preserve dignity and never show unsafe practice as a model. Any needle-insertion illustration must be clearly marked as educational and not permission for self-practice.

17. Scenario-based learning

Scenario 1 — Wrong identity

The request form and wristband do not match. **Action:** Stop, do not collect, resolve through authorised process and report if required.

Scenario 2 — Refusal

The patient refuses after explanation. **Action:** Respect refusal, inform authorised professional, document and do not coerce.

Scenario 3 — Damaged equipment

The needle packet is open. **Action:** Do not use; obtain intact approved equipment.

Scenario 4 — Unsafe request

A caregiver is told to draw blood because the nurse is busy. **Action:** Check scope and supervision; decline if unauthorised.

Scenario 5 — Wrong tube

A test requires citrate but EDTA is used. **Action:** Do not transfer or relabel informally; follow laboratory rejection and incident policy.

Scenario 6 — Dizziness

The patient becomes pale and dizzy. **Action:** Stop, protect, position/monitor according to training, call help and document.

Scenario 7 — Needlestick

A used needle punctures the glove. **Action:** Immediate first aid, urgent report and occupational-health/emergency follow-up.

Scenario 8 — Difficult access

A learner has already made several attempts. **Action:** Stop and escalate; repeated attempts are not uncontrolled practice.

Scenario 9 — Bedside labelling

A collector wants to label at the desk. **Action:** Follow bedside identification/labelling policy.

Scenario 10 — Delayed transport

A sample is left in a warm vehicle. **Action:** Notify laboratory/authorised professional; do not assume stability.

18. Module assessments

Module 1 assessment

1. What is phlebotomy? **Blood collection for laboratory testing.**
2. Why does specimen integrity matter? **Results may otherwise be unreliable or rejected.**

3. Can online completion authorise venepuncture? **No.**
4. What should happen when asked to work outside scope? **Decline and escalate.**
5. Name three purposes of blood testing. **Diagnosis, monitoring and treatment support.**
6. What is the quality principle? **Right patient/specimen/container/label/handling.**

Module 2 assessment

1. Why not rely on bed number? **It is not a reliable patient identifier.**
2. What if identifiers mismatch? **Stop and resolve.**
3. What if the patient refuses? **Respect and escalate.**
4. What does consent require? **Appropriate explanation and voluntary agreement.**
5. How should anxious patients be treated? **Respectfully, with reassurance within competence.**
6. May a caregiver promise results? **No.**

Module 3 assessment

1. Name three vessel types. **Veins, arteries and capillaries.**
2. What must be checked before use? **Packaging, integrity, expiry and policy.**
3. When should sharps be disposed of? **Immediately at point of use.**
4. Should used needles be routinely recapped? **No.**
5. Name three PPE/infection actions. **Hand hygiene, gloves/PPE and single-use equipment.**
6. What follows a needlestick? **Immediate first aid, urgent reporting and follow-up.**

Module 4 assessment

1. State the conceptual sequence. **Prepare, identify, explain, consent, position, hygiene/PPE, site, collect, remove, dispose, label, document, transport.**
2. Does the sequence authorise practice? **No.**
3. Name three stop rules. **Wrong identity, severe pain, dizziness, contamination or distress.**
4. What is a haematoma? **A bruise/swelling from blood leaking into tissue; report and manage within training.**
5. What should happen with nerve-like symptoms? **Stop and escalate.**

6. What is the complication framework? **Stop, protect, immediate measure, escalate, document.**

Module 5 assessment

1. Why do tubes contain additives? **To produce the required specimen type/test condition.**
2. Is tube colour universal? **No.**
3. Why does order of draw matter? **To reduce additive carryover.**
4. What must be checked before publication? **Current laboratory and manufacturer instructions.**
5. Name two capillary principles. **Approved site/device and avoiding excessive squeezing.**
6. What should happen after repeated failed attempts? **Stop and escalate.**

Module 6 assessment

1. Where should bedside labels be applied when required? **At the patient's side.**
2. Name four rejection causes. **Wrong patient, wrong tube, haemolysis, clot, insufficient volume, delay or leakage.**
3. Why document collection? **Continuity, safety, traceability and accountability.**
4. What is the specimen journey? **Collection, identification, labelling, documentation, packaging, storage, transport, reception.**
5. What if transport is delayed? **Notify and follow laboratory instructions.**
6. Should forms go in a contaminated compartment? **No.**

Module 7 assessment

1. Name three special-risk situations. **Anticoagulants, fistula, IV line, infection, bleeding disorder or difficult access.**
2. What is the response to dizziness? **Stop, protect, monitor and escalate.**
3. What follows exposure? **Immediate first aid and urgent reporting.**
4. Does practical endorsement follow online completion automatically? **No.**
5. What does the integrated process protect? **Patient, specimen, staff and laboratory result.**
6. What is the learner's safest response to uncertainty? **Stop and seek authorised guidance.**

19. 40-question final assessment

Recommended pass mark: 80% (32/40). At least 30 questions are scenario/application based.

1. **MCQ:** Phlebotomy is: A) wound dressing; B) blood collection for testing; C) medication prescribing; D) diagnosis. **Answer: B.**
2. **Scenario:** Asked to draw blood without written authorisation. **Answer:** Do not perform; verify scope and escalate.
3. **True/False:** Online completion proves venepuncture competence. **Answer: False.**
4. **Scenario:** Wrong patient label discovered. **Answer:** Stop, quarantine/manage per policy and report.
5. **Application:** State the specimen-quality principle. **Answer:** Right patient, specimen, container, label and handling.
6. **Scenario:** Family requests blood collection at home. **Answer:** Check role, authorisation, supervision and policy.
7. **MCQ:** Blood tests may support: A) diagnosis and monitoring; B) only appearance; C) personal opinion; D) no decisions. **Answer: A.**
8. **Scenario:** Senior says informal practice is acceptable. **Answer:** Request documented scope and supervision.
9. **MCQ:** A reliable identifier is not solely: A) approved identifier; B) bed number; C) matched documentation; D) patient confirmation. **Answer: B.**
10. **Scenario:** Identifiers mismatch. **Answer:** Stop and resolve before collection.
11. **True/False:** Refusal should be respected and escalated appropriately. **Answer: True.**
12. **Scenario:** Anxious patient asks if results will be normal. **Answer:** Reassure without promising; answer within competence.
13. **What should not happen?** Discuss patient details in a public corridor. **Answer:** Protect confidentiality.
14. **Scenario:** Patient needs interpreter/support. **Answer:** Use authorised communication support.
15. **MCQ:** Which is a vein? A) median cubital; B) femur; C) trachea; D) retina. **Answer: A.**
16. **Scenario:** Needle packet is damaged. **Answer:** Do not use; replace.
17. **True/False:** Sharps container should be ready before collection. **Answer: True.**
18. **Scenario:** Colleague asks for recapping. **Answer:** Do not recap; dispose safely at point of use.

19. **Application:** Name four preparation steps. **Answer:** Hand hygiene, PPE, equipment/expiry check, positioning, lighting and sharps readiness.
20. **Scenario:** Needlestick occurs. **Answer:** Immediate first aid, urgent reporting and follow-up.
21. **MCQ:** The venepuncture sequence begins with: A) transport; B) preparation; C) result interpretation; D) diagnosis. **Answer: B.**
22. **Scenario:** Learner plans home practice on a relative. **Answer:** Do not practise on a real person outside approved training.
23. **Scenario:** Patient reports electric-shock sensation. **Answer:** Stop and escalate.
24. **True/False:** Severe pain can be a stop rule. **Answer: True.**
25. **MCQ:** Tube additives matter because they affect: A) specimen type and test condition; B) room number; C) consent; D) identity. **Answer: A.**
26. **Scenario:** Colour memory conflicts with local tube guide. **Answer:** Follow current local guide/manufacturer instructions.
27. **Application:** Why does order of draw matter? **Answer:** To reduce additive carryover.
28. **Scenario:** Several difficult-access attempts have occurred. **Answer:** Stop and escalate according to attempt policy.
29. **True/False:** Excessive squeezing can affect capillary sampling. **Answer: True.**
30. **Scenario:** Child requires sampling. **Answer:** Use age-appropriate communication, equipment, consent and safeguarding procedures.
31. **MCQ:** Bedside labelling means labelling: A) at patient side when policy requires; B) later from memory; C) in a different ward; D) before identity. **Answer: A.**
32. **Scenario:** Specimen is haemolysed or insufficient. **Answer:** Notify/handle according to laboratory policy; do not conceal.
33. **Scenario:** Tube is leaking. **Answer:** Protect, contain and report according to transport/exposure policy.
34. **True/False:** Forms belong in a contaminated specimen compartment. **Answer: False.**
35. **Application:** Name five rejection causes. **Answer:** Wrong patient, wrong tube, haemolysis, clot, insufficient volume, delay, leakage, contamination or storage error.
36. **Scenario:** Patient becomes pale and dizzy. **Answer:** Stop, protect, monitor and escalate.
37. **Scenario:** Patient has a fistula in one limb and IV access in the other. **Answer:** Do not select a site independently; seek authorised direction.
38. **What should not happen?** Continue after refusal because the test is important.

Answer: Respect refusal and escalate.

39. **Scenario:** Transport is delayed beyond local stability time. **Answer:** Notify laboratory/authorised professional and follow instructions.
40. **Integrated scenario:** Wrong request, anxious patient, damaged kit and urgent test. **Answer:** Resolve identity/request, reassure and seek consent, replace equipment, remain within scope and escalate urgency without unsafe shortcuts.

20. Answer key

The answers are embedded in each question. Key reasoning is as follows: safety and scope override pressure; identity and consent precede collection; equipment and infection controls protect patient and collector; technical procedures require supervision; local laboratory instructions override memorised colour sequences; specimen integrity affects the result; complications require stop/protect/escalate/document; and online knowledge must never be represented as invasive-procedure competence.

21. Practical competency verification framework

Practical verification is separate from online completion and must be offered only where the learner's role legally and professionally permits the task. It should use approved equipment, a safe environment, identity verification, a qualified/authorised assessor, current facility SOP and documented remediation.

Station	Expected observed capability
Identification and consent	Matches request, identifiers and patient; explains and obtains consent
Preparation	Performs hand hygiene, PPE, equipment and sharps preparation
Patient positioning and communication	Positions safely, protects dignity and communicates clearly
Venepuncture simulation	Demonstrates approved technique on a training model or under authorised supervision
Tube handling	Selects, fills, mixes and sequences tubes according to current local guide
Capillary simulation	Demonstrates approved device/site principles and disposal

Complication response	Stops, protects, escalates and documents dizziness, bleeding or distress
Labelling	Labels at bedside where required and matches request
Packaging and transport	Uses approved packaging and follows storage/transport instructions
Documentation	Records collection, attempts, complications, handover and incident information objectively

Critical failures: wrong-patient collection; proceeding without consent; unsafe sharps disposal; recapping contrary to policy; continuing after severe pain or neurological symptoms; leaving a patient after fainting; practising without authorisation; mislabelling; concealing a needlestick or specimen error; or signing off an unobserved skill.

22. Downloadable resources

1. Phlebotomy scope-of-practice decision tree.
2. Patient identification and consent checklist.
3. Equipment and preparation checklist.
4. Conceptual venepuncture sequence card.
5. Tube/additive and local-order-of-draw worksheet.
6. Capillary sampling safety card.
7. Specimen labelling and documentation template.
8. Specimen rejection and corrective-action guide.
9. Packaging, storage and transport checklist.
10. Complication response card: Stop → Protect → Immediate measure → Escalate → Document.
11. Needlestick and blood-exposure reporting card.
12. Practical competency assessor checklist.

23. Certificate recommendation

Online CPD certificate

This certificate recognises successful completion of NIC online CPD in the principles of phlebotomy and blood specimen collection, including patient identification, consent, communication, anatomy awareness, equipment, infection prevention, venepuncture principles, tubes and additives, order-of-draw awareness, capillary sampling, specimen handling, labelling, transport, complications, documentation and scope of practice. **This certificate represents online knowledge completion only. It does not establish practical competence to perform venepuncture, capillary sampling or other invasive procedures. It is not a licence to practise phlebotomy, a regulated clinical qualification, professional registration or statutory workplace certification.**

CPD plus practical verification endorsement

This endorsement recognises completion of the NIC online CPD and successful supervised practical verification of the specific skills listed in the attached assessor record on [date], by [assessor/provider], using [equipment] and [current SOP/version]. It represents demonstrated competence only in the specified skills at the time of assessment and does not constitute professional licensure or automatic statutory authorisation.

24. Digital badge recommendation

NIC Phlebotomy & Specimen Collection CPD — Knowledge Completion

Field	Specification
Description	Online CPD knowledge in safe, accurate and professional specimen collection principles
Competencies	Identification, consent, scope, infection prevention, tubes, specimen integrity, labelling, transport, complications and documentation
CPD hours	5–6 online hours
Evidence	Module completion, assessments, final score $\geq 80\%$, scope acknowledgement
Verification	NIC badge URL/QR, certificate ID, date and curriculum version
Limitation	Does not imply practical competence or phlebotomy licence

Practical competency verified endorsement

This may be issued only with a verified assessor record naming the exact skills, assessor/provider, date, equipment, SOP, result, remediation and reassessment date. It must not imply professional registration or statutory authorisation beyond the verified scope.

25. Portal implementation notes

The portal should use:

Course → Modules → Lessons → Knowledge Checks → Module Assessments → Final Assessment → Certificate

Each learner record should distinguish online CPD completion, score, curriculum version, scope acknowledgement, practical verification status, assessor evidence, badge eligibility and expiry/review date.

26. Nigerian context and scope-of-practice notes

Nigerian facilities and communities differ in laboratory access, transport time, staffing, equipment, electricity, referral routes and emergency resources. NIC should require each employer or facility to insert its current laboratory SOP, approved tube guide, order of draw, specimen stability/transport requirements, emergency contacts, exposure pathway and authorised scope before learners practise.

The course should be reviewed by an appropriately qualified Nigerian laboratory, clinical, infection-prevention and governance team. International sources support general principles but do not replace Nigerian law, facility policy, professional standards or manufacturer instructions.

27. References

[1] World Health Organization, WHO Guidelines on Drawing Blood: Best Practices in Phlebotomy.

[2] Centers for Disease Control and Prevention, Collect Adult Blood Culture Sets.

[3] Centers for Disease Control and Prevention, infection prevention and occupational exposure resources for healthcare personnel.

[4] Clinical and Laboratory Standards Institute, Order of Draw for Blood Collection Tubes and Additives.

[5] NCBI Bookshelf, Best Practices in Phlebotomy.

[6] International Organization for Standardization, ISO 15189: Medical laboratories — Requirements for quality and competence.

28. Quality control review

Review area	Finding	Required action
Technical accuracy	Core collection, tube, specimen and complication principles are included.	Laboratory educator must verify every technical detail.
Patient safety	Identity, consent, sharps, exposure and stop rules are prominent.	Add facility-specific emergency contacts.
Scope	Online knowledge is separated from invasive practical competence.	Governance approval before launch.
Nigerian relevance	Infrastructure and local SOP variation are acknowledged.	Insert verified Nigerian facility pathways.
International relevance	WHO, CDC, CLSI and ISO references are provided.	Confirm current versions and access dates.
Assessments	40 questions with scenario/application emphasis.	Pilot and item-analyse before release.
Practical competence	Separate assessor framework and critical failures included.	Approve assessor qualifications and equipment.
Duplication	Complements IPC, First Aid, Vital Signs, Ethics and Documentation CPDs.	Add portal cross-links rather than repeat full lessons.
Accessibility	Plain language, tables, scenarios and diagrams are used.	Test with varied literacy levels.
Visual quality	Visual plan supports understanding without inviting unsafe practice.	Safeguarding and clinical review of images.

Final learner message: “I understand how blood specimens are collected safely, why every step matters, how to protect the patient and specimen, and when I must work under supervision or escalate.”